USA-LUNAR200

Technical data sheet

USA-LUNAR200 is derived from the igneous rock anorthosite, which is composed of >85% calcic plagioclase feldspar. This is derived from the Qaqortorsuaq anorthosite deposit, a near mono-mineralic Archean aged deposit of igneous anorthosite. The chemical composition is very stable with elevated Al and Ca, and a low Si and alkali (K+Na) content.

The mineral anorthite ($CaAl_2Si_2O_8$) is a rare compositional variety of plagioclase and has the highest density of all feldspathic minerals. It also yields the hardest feldspar with the highest refractive index.

The material is bulk handleable with low dust and water content. It can be used in many sectors including as a final product in the mineral wool insulation industry (and others). Lumina's Qaqortorsuaq Anorthosite Mine in Western Greenland has defined resources of over 60 million tonnes and currently holds a 30-year mining license.

Product Development

The data collected and presented in this Technical Data Sheet is based on testing completed by the company. Data presented is for product development and is not intended to be used as a product guarantee. This technical data sheet covers the USA-LUNAR200 product grade; other product specification sheets are available for our 2mm and micronized product range.

Chemical analysis

OXIDE	Wt %	Range
Silicon Dioxide (SiO_2)	49.55	±2.45
Aluminum Oxide (Al_2O_3)	29.80	±2.00
Iron Oxide (Fe_2O_3)	1.00	±0.50
Sodium Oxide (Na_2O)	2.40	±0.25
Potassium Oxide (K_2O)	0.27	±0.20
Magnesium Oxide (MgO)	0.40	±0.30
Calcium Oxide (CaO)	15.40	±3.00
Loss on Ignition (L.O.I. + H_2O)	2.30	±1.15

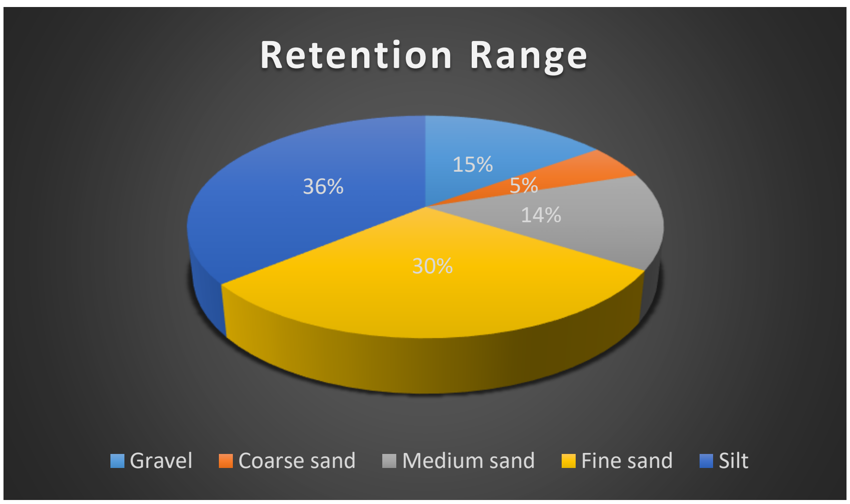
All technical data is averaged results from third-party certified laboratories. This product is still under active development, as such no round-robin analysis has yet been completed. All datasets are ICPMS or lab XRF with internal QAQC checks.

USA-LUNAR200

Technical data sheet

Physical Properties

Powder classification	Diameter range mm	% particle retention range
Gravel	4.75 – 80	13 -17
Sand coarse	2.00 – 4.75	3 -7
Sand medium	0.425 – 2.00	12 - 16
Sand fine	0.074 – 0.42	28 - 32
Silt	Below 0.074	34 - 38



TEST	Value
Hardness (mohs)	6.0 – 6.5
Absorption (H_2O)	0.1 – 0.2 %
Specific Gravity	2.75 g/cm³
Bulk Density, loose pack	2.73 g/cm³
Bulk Density, dry	2.72 g/cm³
Chloride Content	< 0.001 %
Colour	White
Alkali Reactivity	0.006% (±0.002)

*Indicative typical PSD results are shown for information purposes only. Specific physical parameters above.

For additional information see our website
www.lumina.gl or contact Marcos Pineda, marcos@lumina.gl. +52 (614) 191 39 57